Technical Report: Responsible AI Assessment of BrainWaveNet

Responsible AI Assessment for Autism Spectrum Disorder Research

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Abstract

This report assesses BrainWaveNet, a wavelet-based Transformer that classifies autism spectrum disorder (ASD) and typically developing participants using resting-state functional MRI. It combines Leslie's SUM values (Respect, Connect, Protect and Care), FAST principles (Fairness, Accountability, Sustainability and Transparency), and process-based governance across the AI lifecycle. BrainWaveNet is a research prototype evaluated on ABIDE-I; its reported performance and lack of external clinical validation do not support diagnostic use. The assessment identifies evidence gaps in stakeholder participation, subgroup evaluation, test-set governance, reproducibility and operational oversight. It proposes proportionate governance gates, accountable roles and social impact assessment checkpoints. The recommendation is to restrict the model to research until independent validation, meaningful autistic community participation and appropriate clinical safety governance have been demonstrated. It also identifies evidence requirements for reproducibility, human oversight and post-release review if clinical evaluation is proposed. Documented facts are distinguished from recommendations, and remaining limitations are recorded for future work.

Contents

1 Introduction **2**

2 AI Project Background **2**
2.1 Clinical and data context 2
2.2 BrainWaveNet method and reported evidence 3

3 Responsible AI Assessment **4**
3.1 SUM values 4
3.2 FAST principles 5
3.3 PBG landscape, roles and decision gates 6

4 Conclusion **7**

References **7**

1 Introduction

This report evaluates whether BrainWaveNet is suitable for use in a human healthcare environment and examines what would be required for its responsible redevelopment. The project is a deep-learning research model that analyses resting-state functional magnetic resonance imaging (rs-fMRI) data to classify participants into an ASD group or a typically developing control group. The ethical question goes beyond technical classification: we must consider whether the model's purpose, evidence base, design choices and governance justify any real-world use that could affect autistic people. In particular, its binary ASD/control target may simplify a heterogeneous condition and should not be mistaken for a complete account of individual characteristics or support needs.

To answer this, our evaluation applies Leslie's SUM values alongside the FAST principles, translating them directly into a process-based governance (PBG) framework spanning problem formulation, data acquisition, preprocessing, modelling, validation, and deployment. The report distinguishes between information explicitly documented in the published research and issues for which evidence is not available. The absence of information in a paper is treated as an evidence gap, not as proof of deliberate omission. Recommendations are therefore based on the evidence that is currently available.

The scope encompasses this specific model, dataset, and public code implementation, paying close attention to social impact, fairness, accountability, transparency, sustainability, privacy, clinical safety, and technical validity. Our analysis relies strictly on the BrainWaveNet paper, ABIDE-I documentation, and the linked public repository; we do not independently re-run experiments or handle identifiable participant data.

Naturally, this document is neither a clinical trial report, a formal legal opinion, nor a certified medical-device conformity assessment. We avoid inferring individual diagnostic capacity from group-level statistics, nor do we assume that the permissions governing a public research dataset automatically extend to new clinical applications. Where sources omit governance workflows, we log an evidence gap and propose safeguards rather than assuming misconduct. Section 2 outlines the clinical and technical context, Section 3 maps out the SUM/FAST assessment alongside lifecycle controls and decision gates, and Section 4 concludes with concrete recommendations for deployment.

2 AI Project Background

2.1 Clinical and data context

Autism is a heterogeneous neurodevelopmental condition. In clinical practice, identification involves developmental history and behavioural assessment. A neuroimaging classifier should not be treated as a standalone diagnostic tool or as a complete representation of an individual's support needs. Although neuroimaging research has identified group-level patterns of functional connectivity, substantial differences between individuals remain. This limits the extent to which group-level findings can be applied to individual diagnosis (Holiga et al. 2019).

The Autism Brain Imaging Data Exchange (ABIDE-I) aggregates previously collected resting-state fMRI datasets alongside structural MRI scans and phenotypic information. Di Martino et al. (2014) report contributions from 17 sites comprising 1,112 datasets. Their abstract reports 539 individuals with ASD and 573 typically developing controls aged 7 to 64 years, whereas the Results section reports 533 individuals with ASD and 579 controls. This discrepancy in the published group counts should be acknowledged when describing the dataset (Di Martino

et al. 2014; Autism Brain Imaging Data Exchange 2024). The authors also document substantial variation in participant characteristics across sites, including age ranges, intelligence quotient and diagnostic procedures. The sample was predominantly male, and 25% of sites excluded females by design. Multi-site datasets provide opportunities to study larger groups, but differences in scanners, acquisition protocols, recruitment methods and participant characteristics may affect neuroimaging results if not properly addressed (Di Martino et al. 2014; Abraham et al. 2017; Bertò et al. 2022).

2.2 BrainWaveNet method and reported evidence

Jeong et al. (2024) introduce BrainWaveNet to extract spatiotemporal representations from rs-fMRI time series in the frequency domain. Their pipeline applies a complex Morlet continuous wavelet transform (CWT) to blood-oxygen-level-dependent signals. Residual convolutional blocks and frequency-token encoding modules process the resulting complex coefficients. Within each WaveletTF block, the TemporalEnc uses self-attention to learn temporal dependencies separately from the real and imaginary components. The SpatialEnc then applies cross-attention between these components in both directions: the imaginary component is used as the query with the real component as key and value, and vice versa. The model uses a spatial class token for the final binary classification of participants into ASD and typically developing control groups. This describes a group-label classification research task and should not be interpreted as evidence that the model can diagnose individuals.

The authors use CPAC-preprocessed data with a 0.01–0.1 Hz band-pass filter and no global signal regression, and apply the Craddock 200 atlas. The source time series varied in length across sites (78–316 volumes); the authors crop them to 78 time points per subject to obtain a common input length. This means that observations beyond the first 78 time points in longer series are not used by the model. The CWT representation divides the data into five frequency sub-bands, with sampling frequencies adjusted to site-specific repetition times. Following quality-control and exclusion procedures, the study includes 897 participants: 414 in the ASD group and 483 controls. The paper reports a 7:2:1 training, validation and test split and five-fold cross-validation, with model selection based on validation AUC. These are the procedures reported by the authors and are not an independent replication by this project (Jeong et al. 2024).

BrainWaveNet reports an AUC of 0.7388, accuracy of 67.55%, sensitivity of 66.49% and specificity of 68.35% (mean values across the reported folds). In Table 1, it has the highest AUC, accuracy and sensitivity among the compared methods, but not the highest specificity: BrainNetCNN reports 75.32%, BrainNetTF 72.55% and BolT 70.00%. The standard deviations reported for sensitivity and specificity are 9.17 and 9.80 percentage points, respectively, indicating variation across the reported results. These are retrospective research metrics; they do not demonstrate clinical utility, reliable individual risk estimates or improved patient care. The study does not report prospective clinical evaluation or independent external validation on a separate cohort.

Jeong et al. (2024) make their implementation publicly available for inspection, but public code alone does not establish reproducibility. There are specific inconsistencies between the paper's Table 1 and the repository README: BrainNetTF AUC is reported as 0.7141 in the paper and 0.7147 in the README; STAGIN (SERO) accuracy is 57.88% in the paper and 62.66% in the README; and BrainWaveNet accuracy standard deviation is 2.03 in the paper and 2.04 in the README. These appear to be reporting inconsistencies and should be reconciled against the authors' experiment records before claiming an exact reproduction. In addition, inspection of the

official training code shows that test-set metrics are calculated and logged at every epoch, while checkpoint selection is based on validation AUC. This creates a risk that repeated visibility of test results could influence human decisions during development; the code inspection does not establish that test results were used for model selection or that data leakage occurred. A stricter procedure would keep test results inaccessible until model development and selection are complete. The paper's ROI analysis should also be treated as exploratory: attention-based rankings do not, by themselves, establish biological causation, validated biomarkers or individual clinical benefit. Technical pattern recognition must therefore be distinguished from clinical utility and improvements in care. Furthermore, reducing the diversity of autistic people to two classification groups does not capture the full range of their experiences, characteristics or support needs.

The system boundary therefore extends beyond the neural network itself. It includes the MRI scanner, local acquisition protocols, site-specific preprocessing, participant selection, the model's decision threshold, the professional interpreting the output and the institutional framework governing its use. Errors and biases may arise at any stage of this process, rather than only within the model architecture (Suresh and Guttag 2021). This system-level perspective informs the ethical assessment below.

3 Responsible AI Assessment

3.1 SUM values

Leslie's SUM values—Respect, Connect, Protect and Care—provide ethical direction throughout the AI lifecycle (Leslie 2019). They are complementary rather than interchangeable: Respect concerns autonomy and dignity; Connect requires inclusive participation; Protect addresses avoidable harm; and Care concerns wellbeing.

Value	**Evidence and assessment**	**Required response**
Respect	ABIDE-I reports local IRB approval or waiver and anonymisation before sharing, including removal of HIPAA identifiers and facial information. The paper does not specify individual consent scope or all secondary-use conditions.	Verify consent conditions and data-use agreements; restrict access, protect confidentiality and prevent re-identification.
Connect	The paper does not report autistic participation in project decisions. This is an evidence gap, not proof that consultation did not occur.	Involve autistic people in setting priorities, reviewing proposed uses and communicating results; make participation accessible and fairly compensated.
Protect	ABIDE-I is predominantly male; 25% of sites excluded females by design (Di Martino et al. 2014). Representativeness across the autistic population is therefore limited.	Document recruitment and exclusions; test subgroup performance and avoid claims of population-wide generalisability.
Care	Retrospective classification metrics do not demonstrate improved wellbeing. Misclassification could cause distress, stigma, unnecessary investigation or delayed support.	Keep the model research-only; assess harms and ensure outputs never determine access to support (Topol 2019; World Health Organization 2021).

Table 1: Assessment of BrainWaveNet against SUM values.

These values expose a key distinction: data protection alone would not make the system respectful or caring. Even a secure classifier could cause harm if a group-level research prediction were treated as an individual diagnosis. Ethical acceptability therefore depends on purpose, participation and consequences, as well as technical safeguards.

3.2 FAST principles

FAST translates ethical aims into operational requirements. In Leslie's framework, accountability and transparency apply throughout governance, while fairness and sustainability address system properties and wider impacts (Leslie 2019).

Principle	**Evidence gap or risk**	**Required control**
Fairness	Results are aggregate; subgroup performance by age, sex, site or clinical group is not reported. A binary target may also oversimplify a heterogeneous condition (Jeong et al. 2024).	Justify the target and design; report subgroup errors, calibration and uncertainty where sample sizes permit; document limitations and user training (Abraham et al. 2017; Mehrabi et al. 2021; Suresh and Guttag 2021; Rajkomar et al. 2018; Obermeyer et al. 2019).
Accountability	No clinical owner, release authority or incident process is identified. Code logs test metrics each epoch while selecting checkpoints by validation AUC. This risks repeated human exposure to test feedback but does not prove leakage or test-guided selection.	Assign named human owners; log decisions; restrict final test access until development is complete; define who can approve, pause or withdraw the model (Raji et al. 2020; World Health Organization 2021; U.S. Food and Drug Administration et al. 2021).
Sustainability	Energy, hardware and maintenance requirements, and the balance of long-term costs and benefits, are unreported.	Use SIA reviews to assess computational demands, institutional capacity and effects on affected communities; revisit these if the intended use changes (Leslie 2019; World Health Organization 2021).
Transparency	The method and code are public, but paper–README metric discrepancies remain unresolved. Accessible explanations of purpose and limitations are not established.	Reconcile results; version the pipeline; publish a model card explaining intended use, outputs, uncertainty and limitations. State clearly that outputs are research classifications, not diagnoses (Mitchell et al. 2019; World Health Organization 2021; U.S. Food and Drug Administration et al. 2021).

Table 2: Assessment against FAST principles.

The binary ASD/control formulation also raises a question about how autism is represented. Autism is heterogeneous: people may have different characteristics and support needs, so a single group label can conceal meaningful individual variation. Future research could examine validated, dimensional measures of specific autism-related characteristics rather than treating the binary prediction as a complete account of a person. Such measures would not constitute a single "degree of autism" or a diagnosis; clinical interpretation and any diagnostic decision must remain with qualified professionals, informed by the person's history and voice.

FAST principles can conflict in practice. Subgroup reporting may support fairness but create

privacy risks for small groups; transparency must therefore expose methods and decisions without disclosing participant-level data. More validation and monitoring strengthen accountability but require resources, which should be assessed under sustainability. The project should record these trade-offs in its risk register and reconsider them at each SIA gate.

3.3 PBG landscape, roles and decision gates

The proposed process-based governance (PBG) framework assigns responsibility across five lifecycle phases. The sponsor owns purpose and resources; the ML/data lead maintains technical records; the data steward documents provenance and quality; the privacy/security specialist reviews access; an independent reviewer challenges validation; and the clinical safety lead owns the safety case and monitoring. Autistic representatives contribute to priorities and proposed-use reviews. A release board records decisions and outstanding risks.

Lifecycle phase	**Current evidence and gap**	**Control and accountable lead**
Problem formulation	The ASD/control research objective is specified, but no clinical user, care pathway or patient-relevant benefit is established.	Sponsor and autistic advisory group justify the purpose, consider alternatives and prohibited uses, and complete an initial SIA and risk register (Alpha).
Data acquisition	ABIDE-I provides multi-site rs-fMRI and phenotype data; site differences and reuse conditions require documentation.	Data steward records provenance, permissions, quality, missingness and exclusions; privacy specialist verifies access and reuse conditions.
Preprocessing	CPAC, filtering, parcellation, cropping and CWT are described; sensitivity to these choices is not independently established.	ML/data lead versions the pipeline and tests sensitivity to motion, site, cropping, harmonisation and exclusion thresholds.
Modelling and validation	Internal metrics are reported, but subgroup performance and external clinical validation are not demonstrated. Test metrics are logged during training, although leakage is not established.	Independent reviewer freezes the protocol and test set, then assesses discrimination, calibration, uncertainty and subgroup results. Require site-held-out and external validation before release consideration (Alpha-to-Beta).
Deployment and monitoring	No clinical deployment, prospective monitoring or safety process is evidenced; the model remains a research prototype.	Clinical safety lead and board prohibit diagnostic use without a safety case, approvals, training and human oversight. Any future use requires drift/incident monitoring, pause and rollback procedures, and SIA review (Beta).

Table 3: Lifecycle phases, evidence gaps and proportionate PBG controls.

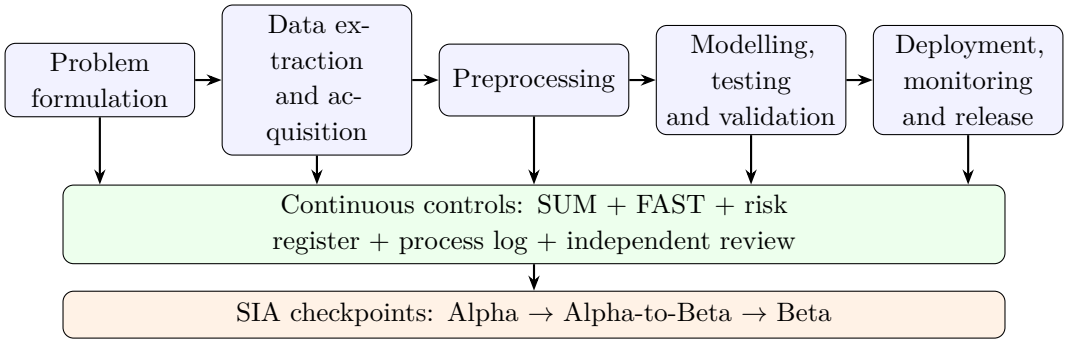


Figure 1: Proposed PBG structure. Ethical review is continuous, with SIA checkpoints at lifecycle transitions.

At Alpha, the sponsor assesses purpose, stakeholders, alternatives and ethical permissibility. At Alpha-to-Beta, after validation, the team revisits the SIA, completes independent review and confirms safety controls before the release board decides. Beta is conditional on future authorised use: the team would review real-world impacts, stakeholder feedback and incidents, and mitigate, pause or withdraw as necessary. Each gate records its decision, approver and outstanding risks, following Leslie's SIA lifecycle guidance (Leslie 2019).

4 Conclusion

BrainWaveNet remains a research prototype: its ABIDE-I results do not justify clinical diagnosis or autonomous decision-making. The reviewed material does not establish autistic co-design, subgroup performance, independent external clinical validation, complete reproducibility or operational monitoring. These are evidence gaps, not proof that harm has occurred. The binary target also limits what the model can represent about individual autistic people; future research could examine validated measures of specific characteristics, but not a single numerical "degree of autism". Such outputs would still require clinical interpretation and must not replace the person's voice or professional assessment.

The appropriate decision is to restrict the model to research and apply the proposed PBG framework across problem formulation, data acquisition, preprocessing, validation and any future release. The initial SIA should be completed at Alpha and revisited before any implementation at Alpha-to-Beta. Beta reassessment is conditional on a future authorised deployment; it is not the project's current status. Clinical use should remain prohibited unless a clear patient-relevant need, meaningful stakeholder support, independent validation, acceptable safety evidence, required approvals and human oversight are demonstrated.

Immediate work should protect the final test set, reconcile paper–repository discrepancies, version the development pipeline, publish a model card and assign accountable owners. Any future pilot must be limited, supervised and reversible, with monitoring for drift, errors, complaints and unintended impacts. Outputs must be identified as research classifications, not diagnoses, and must not determine access to support. If these conditions cannot be met, the responsible decision is not to deploy BrainWaveNet.

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